

AN AFRICAN GOLD NOVEL
THE SPIRITUAL FOURTH

AFRIKA 2100

They were never here for the gold.

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ARJUNA BADGER PRESS

Witwatersrand, west of Randfontein • Landing Day, 2071

She was at the third benchmark by half past four, which was earlier than the job needed and later than she had wanted.

The brass plug was where it had been in 2069, set into a concrete block the size of a bread loaf, stamped **TR/W-114**. Dineo brushed the grit off with her thumb, seated the tripod, and levelled up. Tebogo held the staff at the backsight and complained about the cold in two languages.

“You could have taken leave,” she said.

“Everybody took leave. There’s nobody left to take leave from.”

“Then hold it plumb.”

Overhead the object had not moved in any way a person standing still could measure. It had been up there since she was forty-four. Her sister had cried about it for a week in 2066 and then stopped, and now made jokes about it at Christmas, and this was the general history of the country’s feeling on the matter.

Closure on the morning traverse came out at eleven millimetres over 2.4 kilometres. Acceptable. She wrote it in the field book in pencil, because ink runs and pencil doesn’t, and because her supervisor in 2041 had rapped her knuckles for using a pen and she had never once forgiven him or gone back to ink.

The gravimeter went in at the fourth station, on the pad she had

poured herself in April. Base reading logged 05:12. She would take it again every hour until the thing was on the ground and every hour after that until dark. The Department wanted to know whether the reef moved. Two hundred metres of old stope work lay under this veld, mined out, flooded, and abandoned before her grandmother was born, and if you set anything heavy on the roof of that you might find out what a sinkhole thinks. Nobody in the country had been able to say what the object weighed. The estimates on the radio ran from four hundred tonnes to nine thousand.

“They should have landed it on the Free State,” Tebogo said. “Flat. Solid. Boring.”

“They didn’t ask me.”

“They asked somebody.”

They had, in fact, asked. Coordinates had come down through the mesh in March, in three formats, one of them a format nobody had used since 2038, and the Department had spent five weeks arguing about whether to counter-propose. In the end the Minerals people looked at the old plans and said the ground would hold and the Department said fine, and a fence went up, and Dineo poured her pads.

By seven the radio in the bakkie had gone over to the outside broadcast and she could hear the crowd on the ridge behind her without turning round. Chairs. Somebody’s speaker. A boy shouting the same word over and over for the pleasure of hearing it come back off the mine dump.

The station had put a cricket commentator on it. This turned out to be the correct decision, because he was the only man in Johannesburg trained to describe something arriving slowly for six hours without lying about the pace. He described the light on the underside. He described the fence. He described a woman selling boiled eggs at the cordon and gave her name and the price of the eggs, which had gone up.

“Now, if you’re joining us in the North,” he said, “I’m told you’ve got sound only, so let me do this properly for you. It is a clear morning here on the reef. Wind from the southwest, light. The object is

presently at, they tell me, eleven thousand metres and coming down at about the speed of a lift in a government building.”

Then a professor came on and said the descent was consistent with the profile published in 2066, and had been consistent every day for five years, and that the interesting question was not the descent.

Then a man from Krugersdorp phoned in to read numbers.

He read them well. He had the published telemetry in front of him and he read the ephemeris the way other men read football results: the object’s track back through 2070, 2069, 2068, and Diri’s tapes before that, the whole beautiful boring line of it, and he got as far as the burn.

“August 2061,” he said. “That’s the deceleration onset off the tapes. Peer-reviewed, twice. I’ll say it again because nobody says it. August 2061.”

“And the note was sung in June 2064,” said the presenter warmly, “so we know exactly why they’re here, and I’ll tell you what, my brother, when I was a boy nobody sang for me like that.”

The caller said something about the numbers. The presenter thanked him for the numbers. The next caller wanted to talk about whether it was a public holiday.

Dineo took the 07:12 gravity reading and wrote it down.

The taxis stopped at about half past nine. Nothing announced it. She noticed the sound of the R28 change and looked up and the traffic on it was cars only, and then not even cars. Government instructions had achieved nothing all week. The rank achieved it in an hour, because the drivers wanted to see, and once the drivers wanted to see, the country stopped.

Her nephew Sipho arrived on foot at ten with three litres of water, a bag of oranges, and the news that his mother had bought out the spaza.

“Everything?”

“Mealie meal, tinned pilchards, candles, batteries.” He counted on his fingers. “Sugar.”

“She remembers ’64.”

“She remembers ’65,” he said, which was the worse year, and put the oranges in the shade under the bakkie.

He stayed. He was nineteen and had never known the sky without it. She put him on the levelling staff so that Tebogo could go and eat, and he held it straight, and he watched the object the entire time over the top of the staff, and did not once let it wander.

At 11:04 she took the reading. Nothing. The gravimeter is a drift instrument and its drift that morning was ordinary and dull, and she noted the drift, and she noted the temperature, and she noted that the object’s shadow had reached the fence line on the western side.

It set down at 13:26.

There was no wind from it. That was the thing she had not expected and the thing she would be asked about for the rest of her life. No dust. It came the last thirty metres slower than the last hundred, and the veld under it did not move, and the grass did not lie down, and then it was on the ground with its mass on the ground, and a noise came across the flat to her that she would put in the report as a *low tone, approximately 40 Hz, duration under two seconds, cause undetermined* and which sounded to her, at the time, like a large door in a big empty building being closed by someone being careful.

The crowd on the ridge made no sound for a moment. Then it made an enormous sound, and then it went back to talking.

Dineo watched the gravimeter.

Eleven microgal. Then eight. Then, over four minutes, it settled and sat down at a new number and stayed there. Tilt at the two pads: 3.1 and 2.6 arc seconds, west-down, no rate. The old stopes had taken it. She had expected them to and had not been sure, and she wrote all of it down, station by station, in order, with the times, in pencil.

“Auntie,” Siphosaid. “Is that it?”

“That’s it.”

“Should I put the staff down?”

“Put the staff down. Come and eat.”

Behind them a woman was crying and laughing at once and telling someone on the phone to be quiet and just *listen*, and a man with a portable set had turned it up so that the ridge could hear the studio, where a panel booked five years ago had finally run out of things to say and was reading messages instead.

Sipho ate an orange and said, “They came here.”

“Mm.”

“Not to America. Not to China. Here.” He was grinning. He was not making an argument; he was younger than the approach and he had grown up under it and it had come down on his ground, in his lifetime, on a Thursday. “That has to mean something about us, Auntie. Hey? They chose. That means something about *us*.”

Old Ntate Molefi, two chairs along, said, “It means the coordinates were good.”

“It means something about us,” Sipho said again, happily, and nobody on that ridge argued him down, and the woman with the phone said *amen* absently, still listening to the studio, and somebody further along said that G should have seen this, and somebody else said he did see it, meaning the arithmetic, and both of them were right and neither of them cared to fight about it.

Dineo took the 14:00 reading.

The delegates went in at four.

They had been at the Rand Show grounds for three days in a marquee, arguing about who would speak first, and in the end the order was alphabetical by delegation in French, which pleased nobody and was therefore adopted. The radio carried all of it because the radio carried everything that day. Dineo listened while she re-ran the levelling loop, because the ground had moved and the ground now needed measuring again, and because a person can only look at a thing for so long.

They asked about weapons. Answer: a statement of carried systems by category, three items, with tolerances, and a declared refusal to

discuss capability.

They asked about origin. Answer: a bearing, an approximate transit duration, and a refusal on the location.

They asked whether we are alone. Answer: no.

The ridge went up like a stadium at that and Dineo missed the next four questions.

They asked whether there is a god. Long pause on the tape, which the whole world afterwards agreed was the most human moment of the day and which was, on the recording, four and a half seconds of room noise. Answer: *the question is not one we are competent to hold.*

They asked intentions. Answer: a manifest. A list of quantities of worked gold at stated purities, a schedule, and the word *renewal*, which was in the second sentence and which the translators fought over for a decade.

They asked, again and again, in six languages, whether the note had brought them, and the answer each time was at protocol level and did not say yes, and every delegation in the marquee wrote down yes.

A delegate from the East African bloc asked if they had a name for themselves. Answer: a designation, eleven characters, given twice.

By six o'clock people in the marquee had begun to say *Anunnaki* out loud, first as a joke and then not as a joke, because it was the oldest word anybody had for people who come a long way for gold to mend a dying thing, and the fit made the hair stand up on your arms. The visitors did not accept it. They did not correct it either. They let it sit on the table like a hat somebody else had put there.

Nobody asked what they needed.

Dineo noticed, in the way you notice a missing bolt, and said so to Tebogo while she packed the prism: that when a client comes onto your site the first thing you ask is what he needs and the second thing is what it will cost to do it properly, and that they had done neither in nine hours.

"They asked about weapons," Tebogo said.

"They asked about weapons."

“Well.” He shut the case. “The meat won’t cook itself.”

He was quoting somebody’s uncle from the morning phone-in, and it was already the funniest sentence in the country, and it stayed the funniest sentence in the country for seventy years.

The braai smoke came up off the ridge at dusk and the taxis started running again at about four the next morning, which is to say the day ended the way days end.

Dineo sat in the bakkie with the door open and the field book on her knee and the reading lamp on and worked through the day’s stations in order, transferring them onto the Department’s forms in her small upright hand: TR/W-114 through the loop and back, times, temperatures, closure, the eleven microgal, the tilt at both pads, the 40 Hz tone with *cause undetermined*, the new datum. All of it true. She initialled each page.

Before dawn that morning, an hour before she seated the tripod at the third benchmark, she had walked out past the fence line to a place that was not a station.

Her pencil stopped above the next ruled line.

The lamp buzzed. On the ridge somebody’s speaker was playing music now, and the crowd was thinning, and Sipho was asleep in the tray with his jacket over his face.

She looked at the line for a while. Then she took the pencil away, closed the field book on it, and put the book in the canvas bag and the bag under the seat, and she did not write it down, then or afterwards, for twenty-seven years.

The forms went in on Monday.

By Tuesday she had four requests: a site plan at 1:500 for the cordon, a levelling datum the delegations could all agree to use, a footprint boundary to three decimals for the ledger, and a note from a man in Pretoria asking whether the Department’s monument stamps could be re-cut, since TR/W-114 now sat inside a restricted area and the numbering would have to change.

She re-cut the numbering. It took eleven days.

8WITWATERSRAND, WEST OF RANDFONTEIN · LANDING DAY, 2071

eGoli Spaceport, Berthing Floor Four • six weeks to Renewal

The first thing Naledi did on shift was mistrust her own ear.

The reference bar lived in a steel case bolted to the wall outside the choir line, and the case had a temperature log, and the log had to be initialled, and the bar had to be struck with the wooden mallet and not with a knuckle, no matter how clever your knuckle. She struck it. She held the earpiece off, listened raw, wrote *A-440 ok by ear* in the book, then put the earpiece on and let the counter tell her she was eleven millihertz optimistic.

Eleven. She wrote that down too. Some mornings she was flat, most mornings she was sharp, and in nine years she had never once been right, which was the whole reason for the mallet and the case and the log.

"You're sharp," said Thandeka, coming past with two coffees and giving her neither.

"I'm sharp before eight. After eight I'm a saint."

"You're sharp at fifty. My mother was sharp at eighty. It's genetic and it's ugly."

The floor smelled the way it always smelled: hot platcap bank, cold rock, machine oil, and the sweetish burnt note off the gantry brakes that everybody claimed to hate and nobody ever complained about in writing. Under the smell was the noise, which was not noise once you

had lived in it. Floor Four hummed. Gallery air came up through the grating at eighteen degrees and made a broad grey sound in the low sixties. The e-grav hoists over Bay Two sang thin and high while their rotors were up, and dropped out of hearing when the freight settled. Footfall pavers clicked under everybody's boots, harvesting whatever a person's weight was worth, which was almost nothing, which was the point. Waste no motion. It was written on the wall in the concourse in three languages and a man had been paid too much money to design the lettering.

Naledi loved it here in a way she had given up trying to explain to people at parties.

The choir line was nine stalls in a row against the west wall, each one a lead-lined box the size of a shower, each with its driver ring, its pickups, its little brass door with a proper handle because the guild's founders had come out of harvester engineering and did not trust anything that opened by itself.

Worked gold went in. A reading came out. That was the job, and it took nine years to learn and about forty seconds to describe.

Lot seven of the Kenyan consignment was already sitting in Stall Three under custody seal, thirty-one kilograms, formed to what the manifest called a Class-B ring blank and what everybody on the floor called a doughnut. Naledi broke the seal in front of the Steward on duty, because that was the order — Steward first, engineer second, ledger third — and the Steward wrote the time.

Tefo, that morning. He was her age and he stood like a much older man.

"Morning, ntate."

"Morning." He watched her hands. He always watched her hands. She had decided years ago that this was not suspicion but a kind of attendance, the way her mother watched a pot.

She seated the doughnut on the spider, dogged it down finger-tight and then a quarter turn, and ran the sweep. Twelve seconds of rising drive, the pickups reading back, the counter building its picture. The stall answered. It always sounded to her like a struck glass held under

water: not a note, a *place* where notes went.

Fundamental within four millihertz of the Accord figure. Second and third partials where they should be. Q high. The decay curve came up on the screen and lay down over the reference envelope like a hand into a glove.

“Clean,” she said. “Beautiful, actually. Whoever formed this one wasn’t rushing.”

“Nairobi,” Tefo said.

“Well. Tell Nairobi.” She keyed the sign-off, thumb on the plate, and the ledger took it and gave her back a receipt hash she would never look at and could never alter. *Put it on the ledger and it’s out of your hands* — half the guild’s manuals opened with a line like that, and half the guild recited them like grace before food, which Baba Sipho said was fine as long as they also read the appendices.

Lot eight failed.

Not dramatically. Eight was a plate, eleven kilos, and its fundamental sat nineteen millihertz low with a partial smeared out into a fat little hill that had no business being there. Naledi ran it twice, changed the seating, ran it again, and got the same hill.

Karabo, who was two months out of the academy and would be excellent in about four years, leaned in and said, “Contamination?”

“Sometimes.”

“Silver?”

“Sometimes it’s silver, sometimes it’s a void, sometimes the tolerance on the forming press walked in the night and nobody told anybody.” She flagged it, wrote *reject, partial structure at 2f, recommend section and re-melt*, and let the stall wind down. “Don’t guess in the box. Guess in the canteen.”

“So what is it?”

“I think it’s a void.” She shrugged. “In the canteen.”

Tefo wrote the time on the reject and did not say anything, and the plate went back into custody, and somewhere in Kenya a foreman was going to have a bad Tuesday. That was the trade. The visitors

bought worked gold at stated purity in stated forms to stated tolerances, and they did not haggle, and they did not accept substitutes, and in twenty-seven years no consignment had ever been refused at the pad, because the guild refused it here first. Baba Sipho put it plainly at every induction: *we are the last people who can still say no cheaply.*

The hull for Bay One came in at 11:40, which was 11:40, because it always was.

Naledi went up on the catwalk to watch, along with everyone else who could invent a reason. The gallery mouth is a vitrified bore forty metres across, glassed by something nobody has explained in seventy years of trying, and the catwalk runs straight through it, and she had walked that catwalk so many times that she only sometimes remembered to be amazed. The hull came over the apron without wind. She had heard her grandmother describe that once, on the radio programme they replayed every Landing Day, in the voice of a woman being asked about weather.

Protocol came through on the floor board in the usual form.

BERTH ONE. MASS AT CONTACT 3,410 TONNES. CLEARANCE REQUESTED 4 METRES. HOLD 6 HOURS 40 MINUTES. UPLIFT LOT SCHEDULE 44-C, NINE ITEMS, PURITY AS FILED, FORM AS FILED. TOLERANCE UNCHANGED.

“Tolerance unchanged,” Thandeka read out. “Every time. As if we’d wake up one morning and start rounding.”

“They’re a freight company,” Naledi said. “You’ve met freight companies.”

“I’ve met freight companies who send flowers.”

The liaison’s countersign came a minute later, six lines, signed *Sela*, and one of the lines noted that item four of 44-C had been re-formed since filing and asked that the tolerance be verified against the filed figure rather than the re-formed drawing. Which was correct. Which was, in fact, exactly what Naledi would have asked for in their position, and she had a small private habit of noting when the counterparty was being careful, because on a floor where everybody talked about the

visitors as an event, it was steadying to remember they were also a client with a paperwork culture.

Karabo, chin on the rail: "Do you ever think they're watching us work?"

"They're watching the load," Naledi said. "Which is us working."

She ate outside, on the wall by the east cordon, because the canteen had the sound of eighty people and no ceiling treatment and after five hours of pickups her ears wanted grass.

The fence there is chain-link, four metres, with the pilgrim gate at the end of it, and by noon there were maybe sixty people along it. Ordinary. Some had come for the hull; you could see the hull's upper curve from that stretch, and people held children up. Some had come for the other thing. A woman was pressing a coin against the wire and holding it there with her palm flat, eyes shut, and further down two men had chalked the sentence on the pavement and were arguing about whether the short form counted.

The coins were the ledger coins, the ones with G's face on the obverse, worn smooth around the jaw from exactly this. Naledi had one in a drawer at home from her confirmation, which had been a Fold confirmation on her father's side and a very tense lunch on her mother's.

A young man came along the fence with a clipboard, doing it politely, and asked her whether she worked inside.

"I do."

"Then can I ask you," he said, "what would he have signed?"

He meant the Renewal. He meant it seriously; he had a lanyard and a printed sheet and the patience of somebody who had asked forty people already.

"I don't know," Naledi said. "He died before the Flare. He never saw any of it."

"He saw the arithmetic."

"He saw some of the arithmetic." She wrapped the rest of her bread. "Ask me what the tolerance on item four is. I'll tell you to three places

and you'll be bored."

He wrote something down anyway. She hoped it was *bored*.

Her mother left a voice note at two, which the mesh delivered in a burst when Naledi came out of Stall Six.

Ausi, the man from the Department phoned again about Gogo's field books. I told him the family has them and the family is not a library. Phone your brother, he thinks I was rude.

Fourteen books. Canvas-bound, pencil throughout, initialled at the foot of every page in a small upright hand, and in the margins of four of them a set of bearings and offsets that did not correspond to any monument in the register. Naledi had photographed those pages when she was twenty-three, for reasons she could not now reconstruct, and never shown them to anybody at the port, for reasons she also could not reconstruct.

Gogo Dineo had surveyed the landing ground. Everybody knew that. Every Landing Day somebody played the tape. And she had sat in her chair in Qwaqwa with her ankles swollen and answered every question a granddaughter could think of about datums, closure, and the correct way to hold a staff, and none at all about anything else, and she died in 2098 with her mouth shut and her handwriting intact.

Naledi put the earpiece back in and went to lot four.

Handover at seventeen hundred.

She read the numbers out to Lebo off the book and not off the screen, because that was the discipline: *lots seven through fifteen signed, lot eight rejected for partial structure at 2f, stall three driver ring at nine hundred hours since service, stall six pickup two reading two millihertz proud all afternoon and I've noted it, floor idle logged at fourteen-hundred, uplift 44-C staged, item four verified against filed.*

"Pickup two proud," Lebo said. "Proud how?"

"Proud like a cousin with a new car. Two millihertz. It's in the book."

"You want it changed out?"

“I want it watched.” She signed. Lebo signed. The ledger took both and gave nothing back.

She hung the earpiece on its hook, put the mallet in its clip, initialled the temperature log for the reference bar on the way out, and stood for a second in the doorway with her jacket half on.

The floor was doing what it does in the evening: gallery air coming up grey and broad, one hoist still spinning down over Bay Two, a hull sitting on its pad with its mass on the ground and the room’s idle sitting under all of it, the sound she had worked inside since she was nineteen.

She hummed against it, the way you touch a wall to see if it’s wet.

Then she went up the stairs and out through the concourse where the pavers clicked, and got on the tram, and gave up her seat to a man with a cooler box, and rode home with her fingers going on her knee, still holding the pitch she had hummed, trying it against the pitch she had walked out with. Nothing she could put a number to. Not enough for the book.

Sharp before eight, she thought. Sharp at five, apparently. Genetic and ugly.

She let it go twice on the way home and picked it up again both times.

Lancaster Compact zone, Gordonville depot • six weeks to Renewal

The scale said 908 kilograms and the manifest said 908 kilograms, so Ruth signed the manifest.

That was the third weighing. First at the mint in Philadelphia, second at the rail transfer in Coatesville, third here on the depot platform scale that the Compact used for hog lots and hay and now for this. She had asked for the third weighing herself. Nobody argued. A quartermaster who signs on two weighings is a quartermaster explaining herself to a board in six months.

Twenty-two cases. Each case forty-one kilos of worked gold and eleven of crate. Ash crates, dovetailed, no nails in the corners, made in Intercourse by a shop that mostly built grain wagons. Steel banding. Custody seals on the bands, wire-and-lead, the old kind, because the Accord wanted a seal a man could look at.

She walked the row and put her hand flat on each lid. Not for anything. It is what you do.

“Load’s good, Captain,” Beiler said.

“Load’s good. Wagon.”

They went and looked at the wagon.

It was a gun carriage. It had been a gun carriage in 2066 when the

Army moved field pieces out of Carlisle to sit on the granary roads, and the barrel had gone for scrap in the seventies, and the trail and the axle and the wheels had stayed because nothing built for a gun ever wears out doing anything else. The shop at New Holland had decked it, cross-braced it, hung new sides, and fitted piezo struts in place of the leaf springs, so the ruts on the Blue Ball road would put something back into the cells under the seat.

Two teams of Belgians. Four horses. Sixteen hundred kilos of animal for nine hundred of cargo, which the young men found funny and which Ruth did not, because the animals were the part that would still be working if every cell in the county went flat.

"Reuben says the offside wheeler is short in the shoulder," Beiler said.

"Is he."

"He says it. He's been saying it a week."

"Change him out."

"Reuben doesn't want him changed out. He wants it on the record that he said it."

"Then put it on the ledger and keep the horse." Ruth crouched by the hub and turned the wheel a quarter and listened to the bearing. "Tell Reuben I put it on myself."

Beiler wrote it in his book. He wrote everything in a book. He was forty-four and had been Plain until he was twenty-two and was still Plain in every way that mattered to a load: he did not hurry, he did not shout, and he could tell you the weight of a thing by looking at it and be within five kilos.

The blessing was at seven, before the heat.

Two hundred people came, which was more than she wanted and fewer than she had feared. The depot yard filled up along the fence, and the school let out for it, and somebody's Assist frame stood at the back with its arms folded, and the man in it apologised twice for the noise of the joints, which was nothing, which was less noise than the children.

Bishop's daughter's husband led it. That was how it worked now; the Fold has no ordination and it keeps the shape of the households it grew out of. His name was Eli Hoover and he sold seed.

He stood on the platform scale, which Ruth noticed and liked, because it meant he was being weighed.

He read the folded passage about the corn of wheat. Then the one out of the second book of the Ledger, the ninth kindness, the one about the neighbour's ox and the neighbour's roof, which in the folded text has the ox and the roof and then the sentence, and the sentence is *be kind*.

She had heard him read it before with the whole line in it. Not this morning. Two hundred people and the school children in the front.

The gathering said it back the way a gathering does: half a beat behind, some in Deutsch, some not, the two languages sliding over each other and coming out even.

Sei gut. Be kind.

Ruth said it. She said it with her hat off and her hand on the wagon side, because the wagon was what they were blessing, and she meant it, and she also stood there with the other half of it in her mouth like a stone.

Eli Hoover put his hand on the near case.

"This is not a tribute," he said. "We don't know what it is. We're sending it because we said we would, and because they asked for the work and not the metal, and that's a thing we understand — a man who wants your work is dealing straight with you."

Somebody said amen.

"It might be payment," he said. "It might be a gift. Captain Stoltzfus is going to find out and come back and tell us, and we'll believe her, because she doesn't say things twice."

Ruth kept her eyes on the case bands. That was not the arrangement. The arrangement was that she was quartermaster of a consignment and would deliver it into the port's custody and obtain a receipt. But she did not correct a man in front of his own gathering over a load,

and she noted that she had not corrected him, and she knew she would think about that on the water.

Then the horses were prayed for by name. That is the Plain part and it has never left. Dan, Jake, Sadie, and the offside wheeler, whose name was Hosea and who was short in the shoulder.

The children sang. Two verses and a chorus, nothing anybody in Africa would recognise as a hymn — the tune had come from a hunger-year work song and everybody knew it and nobody knew who made it.

Her mother did not come. Her mother was three miles up the Musser road and eighty-one and had said in April that she would not stand in a yard and watch the Army bless gold.

Her brother did not come either. Levi was fifteen miles away and had been fifteen miles away for twenty-three years. He had stayed; she had gone. He farmed her father's ground and did it better than her father had. They wrote. He signed his letters *your brother Levi* and asked after her back.

She had gone up on Saturday and sat in the kitchen and eaten pie and been asked nothing about the consignment, and had answered nothing, and it had been a good afternoon.

At half past eight the yard emptied and she went round the wagon alone with the checklist on the clipboard, which was paper, because paper does not need a cell.

Bands: twenty-two, seals intact, numbers matched to the manifest. Chocks: in. Brake shoes: clean and dry. Struts: four, charging, needle moving. Cell bank under the seat: at sixty-one percent, which was fine, which was better than fine for a wagon that was going to spend nine days on Compact roads before it saw a railhead. Water: forty litres for the animals, tarpaulined. Feed: eleven bags oats, four bales. Escort: six, mounted, small arms, plus Beiler. Rifles: seven, cleaned last night in front of her. Bible: her grandmother's, German, in the flat pocket of her duffel, where it had been since Fort Lee.

She got to the bottom of the list and stood a minute in the shadow of

the wagon with the yard smelling of horse and dust and the platform scale ticking as it cooled.

She took the coin out of her breast pocket.

It was a ledger coin, worn to a soft brown on the high spots, his face on the obverse rubbed down mostly to the jaw and the eye socket. On the reverse the two sentences, small caps, all the way round the rim, the way the honest ones have them.

DON'T BE A CUNT. BE KIND.

She read it. She read it every time. There was no ceremony in it; it took two seconds; it was the same as putting a hand on a lid.

The soft form is a kindness to children and grandmothers and she had no argument with it. She had watched it happen in her own lifetime — the coins with the word filed off, then the coins struck without it, then the passages read short in company and long alone, and half the gatherings in the county now with the full line on the wall in the back room and *BE KIND* over the door.

It is a warning first. That was all. The first sentence tells you what you will do without trying, and the second tells you what will cost you something. Take the first one out and you are left with a nice idea and no floor under it.

She put the coin back in her pocket and buttoned the pocket.

She was not baptised in. She had been nine weeks from it at eighteen and had gone to the recruiter in Lancaster instead, and no one in her family had ever once made it into a wound in front of her, which was its own kind of pressure. She had a coin and no baptism and a German Bible she could read slower than she could shoot.

Beiler brought her the road order at nine, and the weather off the mesh, and a peach.

"Ninety-one today," he said. "Ninety-four Thursday."

"We'll water at Blue Ball and Gap."

"Reuben says Gap has the trough out for repair."

"Then Reuben knows where the next water is and he'll tell me at Blue Ball." She ate the peach. It was a good peach. "Amos."

"Captain."

"When we get there. If I'm asked to stand up and say what this is."

He waited. He was good at that.

"I don't know what it is," she said.

"No."

"Eli said I'd find out."

"Eli sells seed," Beiler said. "He also sells hope, and he doesn't charge for it, so it comes out about even." He put the peach stone in his pocket rather than the yard. "You'll say what you saw."

"That's the whole of it?"

"That's the whole of it. You've never been any good at the other."

She walked the row of cases one last time, twenty-two lids, hand flat on each.

Then she climbed up and took the seat on the left because Reuben drove and a quartermaster does not drive, and Reuben said something to the near team, low, in Deutsch, that was not an order and not a prayer.

The chocks came out. The wheelers took the weight; the leaders leaned into it; the trail creaked at the pin and the whole nine hundred kilos came off the platform apron and onto the gravel, and the needles on the strut gauges jumped as the first rut took the offside wheel.

Somebody's dog came out of the depot office and ran the length of the fence with them, barking, and gave up at the gate.

They went out the Gordonville road at four miles an hour, east into the sun, with the escort at a walk behind and the dust standing up in the ditch grass and not settling until they were past the schoolhouse.

eGoli Spaceport, Berthing Floor Four • six weeks to Renewal

The discrepancy came out of a housekeeping job nobody wanted.

Uplift 44-C had gone out on the Thursday hull and the guild's part of it was closed, but the Accord's annexes require the tuning house to reconcile each uplift against the counterparty's stated use, once, within seven days, and file the reconciliation. It is a paper exercise. The visitors file a repair manifest for every consignment: item, form, mass, and a use-line saying which component the metal is for. The guild files back a resonance sheet saying what each item measured. The two go into the ledger side by side and nobody reads them again, ever, in the whole history of the trade.

Naledi drew it because she had lost a bet about a rugby match.

She took the terminal in the tuning office, which is a converted store cupboard with a window onto the gallery mouth and a kettle that trips the ring main if anyone runs the desk lamp at the same time. She had the lamp off. She had 44-C's nine resonance sheets on the left of the screen and the manifest on the right, and a mug going cold at her elbow, and she was going to be finished by four.

Item one. Ring blank, 62.4 kg, use-line: *guide segment, ring assembly, class 3*. Sheet: fundamental 212.113 Hz, partials clean, Q high.

Item two. Plate. Use-line: *coupling face, secondary*. Sheet: 388.006.

Item three, four, five. Fine.

Item six was a ring blank too. Same class, same nominal mass within a kilo, same use-line word for word: *guide segment, ring assembly, class 3*.

Fundamental 212.098.

Naledi looked at that for a while.

Fifteen millihertz is nothing. Fifteen millihertz is her own ear before eight in the morning. Two ring blanks formed to the same drawing in two different shops on two different presses will differ by more than that, and the tolerance band in the annex is generous by design, because the visitors buy to a tolerance and not to a number, and the guild's whole existence is the width of that band.

But the guild does not read the band. The guild reads the *shape*.

She pulled up the two decay envelopes and put them one over the other. Item one's third partial came down like a dropped stone. Item six's third partial came down and then sat, for about four hundred milliseconds, on a shelf.

She had seen that shelf before. Everybody had. It is what a driven mass does when it is coupled to something that wants to keep ringing at a slightly different pitch — a stall with a loose spider, a plate that has been seated against the lead lining, a pickup two millihertz proud. It is a coupling artefact and the correct response is to reseal and rerun, and if it recurs, service the stall.

Item six had been read in Stall Five. Item one in Stall Five. Same stall, same morning, same engineer — Lebo, who was better at this than Naledi and had signed both without a note.

She got up, went out onto the floor, and reseated nothing, because both items were four hundred and eleven kilometres up and out.

So she did it the long way, which took nine days and mostly happened between other things.

She pulled every reconciliation for every consignment of the last three years. Two hundred and sixty-one uplifts, one thousand nine

hundred and forty items. The ledger holds them all and the ledger is not slow; the slow part is that the use-lines are free text.

The visitors write their manifests in protocol English and their protocol English is exact but it is not a database schema. *Guide segment, ring assembly, class 3. Guide segment (ring), class three. Ring assembly guide, cl.3, replacement.* Naledi spent two evenings building a table by hand to collapse the variants, and she built it in pencil first, on the back of a stall service log, because that was faster than arguing with software.

Then she sorted every item in the three years by use-line and looked at the resonance sheets grouped that way.

The class-3 guide segments came out in two families.

Not a spread. Two clumps, with a gap between them you could park a hull in. Family A sat around 212.113 and had clean fast decay. Family B sat fifteen to nineteen millihertz low, always low, never high, and every single one of them had the shelf on the third partial.

Two hundred and six class-3 guide segments in three years. Ninety-one in family B.

She sat back and the chair squeaked and she said, out loud, to the kettle, "Oh, that's not nice."

Because a use-line is a *use*. It says what the metal is for. And if the counterparty is buying two acoustically distinct components and filing both under one use-line, then either they have a sloppy clerk, or the two families go to two different places, and the manifest is not describing the second place.

She checked the sloppy clerk hypothesis, because you do. She sorted family B by filing date, by forming shop, by mass, by purity, by the shipping hull, by the day of the week. Family B came from Kenya and Australia and the Bushveld and Chile and one lot out of Aksum. Family B included items formed to the drawing by four presses that had never made anything else. Family B ran all three years with no trend, roughly two in five, and the split held to within a couple of percent across every quarter she cut.

Nobody's clerk is that consistent. Consistency is the fingerprint of a system.

She did the last check on the ninth evening with the lamp off and the kettle unplugged. She took the twelve family-B items with the most complete sheets and looked not at the fundamental but at the interval between family A and family B: 212.113 against 212.098, 388.006 against 387.978, 145.220 against 145.209.

She worked the ratios on paper. They came out the same to four places. Same interval, every pair, every form, three years. Family B was not a family of flawed parts. Family B was family A transposed — the whole set shifted down by one fixed, tiny, deliberate amount.

You do not get that from voids. You do not get that from silver. You get that from somebody forming metal to a *different reference pitch* and calling it the same component.

Naledi's stomach did the thing it does on a hoist.

She wrote it up properly. She had it in her hands by eleven that night and she did not send it, because at eleven at night everybody is a genius. She read it again at half past six with the reference bar log in front of her and her ear eleven millihertz optimistic as usual, and it still stood up.

Discrepancy report. Guild form 12. Observation, method, data appendix, and one recommendation: *that the tuning house request clarification from the liaison as to whether use-line "guide segment, ring assembly, class 3" covers two distinct component specifications, and if so, that the annex reconciliation format be amended to distinguish them.*

That was all she asked for. A clarification and a form change. She was rather pleased with how boring she had made it.

She filed it Tuesday, 08:14, and the ledger gave her back a receipt hash, and she went and did lots eleven through nineteen.

The file came back on Friday.

STATUS: CLOSED. DISPOSITION: NO ACTION REQUIRED. NOTE: VARIANCE WITHIN ANNEX TOLERANCE. RECONCILIATION FORMAT UNDER REVIEW BY COMMISSION AHEAD OF RENEWAL; SUBMISSIONS SUSPENDED.

Time between filing and closure: seventy-nine hours, which included a weekend.

She stood in the corridor outside Stall Two with her earpiece around her neck and read it three times and could not find the part that was wrong, which was the annoying bit. Every sentence was true. The variance was within tolerance. The format was under review; she'd had the circular in March and deleted it.

"Two millihertz proud," she said to nobody.

Karabo went past with a spider wrench and said, "Sorry?"

"Nothing. Get me the service book for Five."

Stall Five's driver ring had been changed in February. Pickups replaced last August, both. Calibration against the bar within four millihertz every week for fourteen months, initialled, dated, temperature logged. The stall was not sick. Naledi had known that before she asked. She asked because a person who files a report and then does not check her own instruments is a person who will be asked why she did not check her own instruments.

Marais came down to Floor Four on the Wednesday, which was not remarkable. He came down most months. He had come out of the guild himself, forty years ago, and he still put on the earpiece and asked to hear the idle, and the floor liked him for it and the floor was not wrong to.

He asked her to walk with him on the catwalk. He was seventy-one and he walked slowly through the bore and stopped at the middle, where the glass gets that green depth in it, and put his hand on the rail.

"You know what my father did on this apron," he said. "He drove a water truck. Nineteen years."

"Ntate, I know."

"I tell everyone. It's the only story I have." He looked down the length of the bore. "Your report was good. I want to say that first, and I want you to believe I mean it, because in about four minutes you're going to think I'm managing you. It was good. The interval work especially

— you didn't chase the millihertz, you chased the ratio. That's the guild at its best and I read it twice."

"Thank you."

"It's also correct, as far as I can tell." He said it plainly, the way you'd read out a mass at contact. "I don't know what those ninety-one items are. Neither do you. Neither, I suspect, does anyone in this building. Now let me tell you what week it is."

He let go of the rail and turned to face her.

"Six weeks. The Renewal instrument is a hundred and nine pages and the volume schedule alone took four years to agree, because volume is the only clause in it that the other twenty-two signatory blocs actually fight over. China wants doubled tonnage and a shortened cycle. The Southern Carry wants the cycle held and the technique schedule front-loaded. The North American Compact wants proof it isn't paying rent. And every one of them has domestic weather that would eat their delegation alive if they came home with less than they promised."

"I know."

"You know it the way you know the weather in Chile. Let me be specific, because specific is the only thing you and I both trust." He counted on his fingers, no theatre in it. "The platinum framework is annexed to the Renewal. Not adjacent. Annexed. If the Accord lapses without instrument, forty percent of the world's platcap electrode supply falls back to bilateral contracts negotiated in a panic, and the price of a cell in Nairobi triples inside a quarter, and every mesh node in the Congo basin that runs on a two-hundred-rand cell goes dark by degrees. Second. The technique schedule. There are eleven deliveries queued behind signature, one of which is a ceramic process for piezo harvesters that halves the cost of a footfall paver. That is not glamour. That is a village in Limpopo lighting a clinic. Third, and this is the one I lose sleep on. This country has been indispensable for twenty-nine years. Do you know how long any country has ever been indispensable?"

"No."

"Neither does anyone, because nobody has been. We are hosting the species relationship on our ground, under our custody, on our terms,

and we are doing it because the Accord says so. Instruments say so. Take the instrument away and what's left is that we happen to be sitting on the gold, and the world has a long and well-documented history of arrangements for people sitting on gold." He put his hands in his pockets. "My father drove a water truck on an apron owned by other people. I would like to die before that comes back."

A hoist came up over Bay Two and the rotors sang thin and then dropped away.

"So," Naledi said. "Drop it."

"No. Hold it." He shook his head slightly. "I'm not asking you to unsee it. I'd fire you if you could. I'm asking you to be curious in September. Curiosity has a season, Naledi, and the season is the twenty-seven years we've had and the fifty we'll have after signature. We have been allowed to be curious for twenty-seven years. We get to be sovereign for six weeks. If I take a question about ninety-one guide segments onto that floor now, I don't get an answer. I get a delegation from the Give bloc saying the host nation is inventing doubt to raise the price, and a delegation from the Keep bloc saying they *told us* the metal was going somewhere unstated, and eleven technique deliveries in a drawer, and the clinic in Limpopo not lit. And I still don't have an answer, because the counterparty will file six lines and sign them and the six lines will be true."

"They usually are."

"They usually are." Something moved at the corner of his mouth. "Re-open it in September. I'll co-sign the request myself, and I'll put my name above yours on it so it isn't a junior engineer against a protocol, and I'll make sure the format review takes your amendment. That's not a bribe. That's a plan, and it's a better plan than the one where you're right in August and unemployed in October."

He waited. He was good at waiting; he'd been in rooms.

"Understood," Naledi said.

"That's not agreement."

"No, ntate. It's understanding. You asked for one and I've got it."

He laughed, actually laughed, and patted the rail twice, and went off

down the catwalk with his hand out for the rail every third step, because the floor grating is open there and he was seventy-one.

She stayed in the bore a while.

She could not tell him the real thing, which was smaller than the country and larger than a form change. Lebo had signed both sheets. Lebo, who had eighteen years on her and had taught her how to hold the mallet, had put her thumb on the plate twice in one morning and certified two items as the same component, and they were not the same component, and Naledi had let a receipt hash go into the ledger under her own thumb ninety-one times over three years without seeing it.

She had written *A-440 ok by ear* every morning of her working life and then let a machine correct her, on purpose, because she did not trust her ear, and the whole point of that discipline — the mallet, the case, the temperature log, the humiliation at eleven millihertz — was so that nothing would ever get past the house. Not *nothing important*. Nothing.

Ninety-one items had got past. Twice a week, for three years, past everybody, into the ledger, out through the bore, up.

The gallery air came up grey and broad through the grating.

She went down to the tuning office and reopened her pencil table, and started a second column, and headed it with the interval — the actual number, four places — because if family B was formed to another reference, then somewhere there was a bar somebody had struck to get it, and she was going to find out whose mallet.

Brixton, her mother's house

• five weeks to Renewal

The survey books were in the second bedroom, in a tea chest, under a duvet cover nobody had thrown out.

Nine books. Buff card covers, cloth spines, the kind the Council issued until about 2080 and then stopped issuing, so Gogo had bought the last four out of a stationer in Fordsburg and complained about the paper for eighteen years.

Naledi's mother brought her tea and stood in the doorway with her arms folded.

"You're looking for what."

"Ground. There's a gallery mouth in the north-west quadrant that the port's plan says was capped in '74 and the walk-through says was capped in '81."

"And your grandmother is going to settle it."

"Her books ran from '69. Nobody else's do."

Her mother made a sound and went back to the kitchen. She had spent two years not sorting this room and was not going to be hurried into it by a daughter with a question.

The books are not diaries. Nobody who has read one would call them that.

Left page: station number, coordinates, instrument height, date, temperature, the initials of whoever held the staff. Right page: the sketch,

and the sketch is to scale, and where it is not to scale it says so, in capitals, with the fudge factor written next to it. Book Three had a whole leaf of tension corrections for a chain that had been dropped down a shaft and straightened by a man in Roodepoort who charged too much. Gogo had written the man's name and what he charged.

Naledi worked through them at the desk with the window open. Traffic on Fulham, someone's radio, the pigeon that lived on the sill and had opinions.

The pencil was 4H everywhere. Hard, grey, thin. It does not smudge and it does not forgive.

Book Five, 2076, a fortnight of levelling along the Kimberley reef line. Gogo had noted, on the third day, *staff-man late, drunk, not again* and then on the fourth day the same man's initials and no comment, and then on the ninth day, *P. Motaung's back — sent him home, did staff myself, closure 4 mm over 1.9 km, acceptable*. Naledi had never heard of P. Motaung. He had a back and a bad week in 2076 and here he was.

There were tram tickets used as bookmarks. There was a receipt for a tyre. In Book Seven a child's drawing of a horse had been folded flat and left at the page for station 44, and the horse had five legs, and on the back in ballpoint someone had written *NALEDI 4* in a hand that was not Gogo's.

She looked at that for a while and then put it back where it was and turned the page.

Book Two was 2071.

She had expected to find the landing in it. There was nothing about the landing. There was station 118 on the eleventh and station 119 on the fifteenth and between them four days of no stations, and a note on the left page, in the temperature column, that said *hot*.

That was very Gogo. Her mother had told a story about it — the whole country on the roofs and Dineo Mokoena on the reef with a level, and afterwards, when the family asked her, she had said the admin took longer than the ship.

Book Two, station 131, was the north-west quadrant. Bearings, chain

distances, the vent shaft, the two survey pegs, the collar of the old Number Six ventilation adit with its year of sinking. Everything closed. Everything in the right column.

The sketch on the right page ran a long spur south-east from the collar, dog-legging twice, the way an adit does when it is following something and not going somewhere.

The spur stopped mid-page.

Not at a margin, not at a fold, not at the end of a line. It stopped in white paper with a good two inches of clean sheet below it and no station number, no coordinates, no *ends*, no *fallen*, no *not entered*, no fudge factor in capitals. The 4H line came down and stopped and there was a dot where the pencil had been lifted, slightly heavier than the line, the way a hard pencil leaves a dot when the hand stays a moment after it has decided.

Naledi turned to the back of the book where the station index was, in Gogo's handwriting, all the way through, all nine books, every number accounted for.

131. 132.133.

She turned back. She counted the pages. She checked the front and rear pastedowns for a knife line, because the honest thing to check first is whether a page has been cut out, and no page had been cut out; the signatures were sound and the sewing was original.

She held it up to the window and turned it, in case there was an impression from a sheet written on top of it. There was the ghost of station 132 coming through from the other side, and nothing else.

She looked at the spur again. Sixty metres of it, drawn to scale, bearing 138 degrees magnetic, with the two dog-legs sketched at the angles they actually were, because Gogo did not straighten things to make a drawing tidy.

Then white paper.

Her grandmother had put a man's name in a book for overcharging on a chain. She had recorded a staff-man's back.

Naledi sat with the book open on her knees and her tea gone cold on the desk and did not know what she was looking at, and knew exactly

how much of it there was.

"Ma. When she stopped surveying. Was it her eyes?"

"It was 2081 and she was sixty-four and the Council put everything on the machines." Her mother was washing a pot she had already washed. "She said the machines were fine. She said she wasn't going to hold a staff for a boy with a tablet."

"Did she keep going down?"

"Down where?"

"Underground. After."

Her mother stopped with the pot in her hand. "She went to the funerals. Miners' funerals. Every one she heard about, until she couldn't walk to the taxi." She set the pot on the rack. "I asked her once why she went to strangers' funerals and she said they weren't strangers."

"That's not an answer."

"It was hers."

She took Book Two and left the other eight in the chest, and her mother watched her do it and said, "Bring it back."

"I will."

"Not in a year. Bring it back."

At the gate the pigeon came off the sill and went over the roof and the radio next door was doing the Renewal, some panel, a man saying the word *sovereign* twice in a sentence.

On the taxi Naledi put the book flat on her knees and opened it to station 131 again and photographed the page with her phone, badly, twice, and then a third time square.

Then she sat with her thumb on the white part of the sheet, which told her nothing, because paper does not tell you anything. She wrote in her own notebook: *NW quad, adit 6 collar, spur bearing 138M, 60m + ?*

She looked at the question mark. Then she wrote the bearing again, in the margin, so it would be on two pages.

The keeping house, Langlaagte • five weeks to Renewal

The Stewards do not have an office. They have a house, and you cannot phone it.

What you do is put a request in the custody box at Gate Nine, on paper, and a Steward reads it, and if the answer is yes somebody tells you at your work. Naledi wrote hers on a stall service log because she had nothing else in her bag: *Tuning house, N. Mokoena. Question about the north-west quadrant and about the 2064 keeping roster. I can come any day after four.* She folded it and put it in the slot and felt like a child posting a letter to a farm.

Four days. Then Karabo, at the tea urn, with the air of a man handing over contraband: “Somebody left this at the desk for you. It says Thursday.”

Langlaagte on a Thursday afternoon smells of dust and cut grass and the burnt-sugar coming off the reclamation plant. The fence runs the whole way along Main Reef, three metres of it, mesh with the port’s green stripe woven through, and along the fence there are people, because there are always people.

The pilgrims had the shade side. Nine or ten of them, a folding table, a paper icon of a face everybody knows, printed cheap and creased at the eyes. A woman was pressing coins into palms as you passed. Naledi took one because refusing costs more time than accepting.

Brass, thumb-sized, warm from the woman's hand. G on one face, in the profile the century has settled on, which nobody who knew him thought looked like him. On the other face, two words, stamped shallow: *Be kind*.

"What would he have signed, sister?" the woman said. Not aggressive. She had a stack of the icons and a bucket for the printing costs and a child asleep on a blanket under the table. "That's all we're asking. Volume doubled, cycle shortened. What would he have signed?"

"I don't know."

"He'd have asked who was in the room." She said it comfortably, the way you say a thing you have said four hundred times and still believe. "That's all. That's the whole devotion."

Forty metres on, on the sun side, the other lot had a trestle and a solar printer and a paper spike, and a man in a Codehouse T-shirt was reading numbers off a sheet at conversational volume, over and over, to nobody in particular.

"Uplift forty-four C. Nine items. Ledger receipt ends four-seven-two-D. Payment schedule entry, page eleven of the technique annex, published, dated, hashed. If a man tells you the tonnage is secret, ask him for the hash. Just the hash. He won't have it."

He caught her eye and held up the sheet, hopeful.

"I know that one," Naledi said.

"Then you know more than the panel on the radio." He was young and sunburnt and pleased. "No offence to the coin people. Their heart is fine. But you can't audit a face."

The pilgrim woman, up the fence, was singing something now with two others, low, in Sesotho, and it was not a hymn Naledi knew, and it was good.

The keeping house is a mine manager's house from 1934 with the veranda glassed in. There is a lemon tree, a Toyota under a tarpaulin, and a brass plate by the door with nothing on it but a number.

Ma-Rethabile Khumalo answered with her sleeves pushed up. Eighty-one, short, hands like a woman who had done something with them

for sixty years, which she had: forty of it on the winders at Number Four.

“Mokoena,” she said. “The tuning house. You wrote on a service log.”

“I didn’t have paper.”

“Good.” She stood back. “Come. I’m doing the candles and I’m not stopping.”

The room off the passage had a long table, and on the table were candle stubs in a bread crate, and a saucepan, and a stack of new wicks on card. Ma-Rethabile sat and went back to it: melt, pour, set, trim. The dip-hooks were on a wooden rack blackened at the tips. A young man was at the other end of the table with a ledger tablet and a face like a man waiting to be introduced.

“Tefo,” he said, and shook her hand across the crate. “I’ve read your form twelve.”

Naledi put her bag down. “It’s closed.”

“I know. That’s why I read it.”

“Sit,” said Ma-Rethabile. “Hold this.”

So Naledi sat and held the mould steady while the old woman poured, and the wax came up the tube and the smell was clean and animal, and she found she had been talking for four minutes about ninety-one guide segments before she had planned to say anything at all.

Ma-Rethabile poured a second. “Trim those three.”

Naledi trimmed the three.

“Your grandmother was Dineo.”

“Yes, mma.”

“I met her twice. She wouldn’t take tea.” The old woman set down the saucepan. “Bring the light.”

The room at the back had been a pantry. It had a slate floor, no window, and shelving up to the ceiling, and it was cold the way rooms are cold when they are on ground and not on foundations.

Ma-Rethabile did not explain the room. She took things off shelves and put them on the slate.

A drum-recorder chart, unrolled and weighted at the corners with two spanners and a stone. Twelve hours of it, pen-ink on waxed paper, the trace flat and thin for the first third and then, from a point marked in pencil, wide — a band, not a line, the pen thrown so hard it had torn through in three places. Along the bottom in a clerk's hand: date, 14 March 2064, and the hours, and *Coupling Gallery, No. 2 pen, drum speed 1 rev/hr.*

A roster. Foolscap, ruled by hand. Names down the left, four columns to the right: down, up, hours, *voice*. Forty-one names. Beside eleven of them, in the last column, a tick. Beside three, a dash. Beside one, nothing at all and a bracket drawn around the whole line in a different ink.

A pair of shoes, size six, women's, the soles worn through at the ball of the foot and the uppers stiff with something pale.

A tally book of the sort a shift boss uses, open at a page of five-bar gates.

A tin of Vaseline.

"That's the fourteenth," Ma-Rethabile said, of the chart, and nothing else. She moved the stone two inches so the paper lay flatter.

Naledi crouched. She could read a trace. She had been reading traces since she was nineteen. She could see the pen going wide and she could see, in the wide part, the periodicity — regular, coarse, about a two-hour beat — and she could see it damp out and go wide again, four times, and then, in the last hour of the drum, the band narrowing to a line as thin as the start and staying there.

"What was the drive," she said.

"I don't know what you mean by drive."

"What was making it move."

Ma-Rethabile picked up the roster and set it down closer to Naledi's hand. "Forty-one people went down. Eleven of them were singing. Three had been singing and stopped."

"That's not a mechanism."

“No.”

“Mma, I’m an engineer. Two hours of beat at that amplitude is energy from somewhere, and eleven people in a gallery is not—”

“I know what eleven people is.” The old woman’s voice did not go up. “I was one of them.”

Naledi shut her mouth.

“You asked about the roster. That is the roster.” She turned it a quarter-turn, squared it to the edge of the slate, and left it. “The rest is sealed and I did not seal it and I cannot open it. The Court holds one part and the Accord holds one part and I hold this part, which is the part that is objects, and objects I can put on a table.”

On the shelf above, behind a folded blanket, there was a wooden case the size of a bread tin with a wax seal on the hasp gone brown and a Court sigil pressed into it. Naledi looked at it for exactly as long as it took Ma-Rethabile to notice her looking.

“Yes,” the old woman said. “That one stays shut.”

Tefo had followed them in with the tablet under his arm.

“Ask her the other thing,” he said.

“Ask me what.”

“Whether she’ll put any of this on the ledger.” To Ma-Rethabile: “Mma, she came to us. She’s guild. She’s already been closed once. If we’re ever going to state a reason, this is the—”

“Not in the pantry, Tefo.”

“Then in the kitchen, but let me say it in front of her, because I’m tired of saying it to you alone.” He turned to Naledi and he was not performing; he had the flat delivery of a man on a loop. “Two consignments halted in twenty-seven years. No reasons given. Not to the Commission, not to the counterparty, not on the ledger. Do you know what that looks like from outside this house?”

“Mysticism,” Naledi said. “Or money.”

"Mysticism or money. Thank you. Half the port thinks we hear voices and the other half thinks somebody paid us, and the Firstborn have decided we are the chosen keepers of the sacred blood, which we have refused four times in writing and they read the refusal as modesty." He tapped the tablet against his leg. "A reason on the ledger can be checked. It can be argued with. It can be wrong and corrected. A silence can only be interpreted, and everyone interprets it in their own favour, and we don't get a vote. Mma, we are handing the movement a relic. Us. The order that was miners first."

Ma-Rethabile rolled the drum chart, slowly, from the far end, keeping the waxed paper off the torn places.

"In seventy-eight," she said, "we halted a consignment. Ninety kilograms, formed, ready. We gave no reason. Within a month there were eleven people who had been in the gallery that year whose names were being read on the radio, and one of them lost her house, and another was met in a car park in Benoni by two men who wanted to know what she had heard." She slid the roll into its sleeve. "If I had put my reasons on the ledger, my reasons would have contained the reason we halted, and the reason we halted was a thing three people said, and I would have had to name the three people or write a sentence nobody could check. You want a reason that can be argued with. So do they. That is what an argument is for — you can win it. Sometimes what I have is not strong enough to win and it is still true."

"So we keep it because it's weak."

"We keep it because it is not ours." She put the sleeve back on the shelf, level with the others. "You will get your ledger, Tefo. You will get it while I am still alive, probably, and I will sign it and I will not be sorry. But you must be able to say the reason in one sentence and mean every word, and at Renewal, right now, you cannot. Neither can I."

Tefo looked at the shelf for a while. "That's a better answer than the one you gave me in June."

"I've had since June."

At the passage door, going out, Naledi stopped.

There was a survey sketch pinned to the wall by the coat hooks, unframed, the paper gone the colour of weak tea. No station number, no coordinates, no title block. A collar, a shaft, a spur running off at a shallow angle with two dog-legs in it, and a scale bar drawn in by hand with the divisions stepped off by dividers.

4H pencil. Hard, thin, grey. Where the spur ended there was no annotation, and the line stopped in clean paper with a dot at its end, slightly heavier, where a hand had stayed a moment after it had decided.

“Where did this come from,” Naledi said.

“A box,” Ma-Rethabile said. “In the winter of '64. Left at the gate at Number Four with the tally books. Whoever brought it did not stay.”

“It’s not signed.”

“No.”

Naledi looked at it a while longer. The bearing was not written anywhere on the sheet. She had to hold her hand up and estimate it off the north arrow, and she made it about a hundred and forty degrees, and she knew perfectly well how badly you can lie to yourself with a hand held up against a wall.

She did not photograph it. She could not have said why.

“Come back when you know what you want,” Ma-Rethabile said at the door. “Not when you know what you think.”

On the fence walk back the pilgrims had grown to twenty and the printer man had gone home, leaving his paper spike on the trestle with the sheets still on it, weighted with a brick, for anyone.

Naledi turned her phone on at the taxi rank and it gave her fourteen days of nothing and then one thing.

ACCORD COMPLIANCE OFFICE — COURTESY NOTIFICATION. *Ref: Renewal secretariat audit trail, cycle 3. Your ledger read-queries for the period 12 July – 2 August have been aggregated into the pre-signature audit bundle in accordance with annex 9(4). Queries listed: 2,114. Categories: reconciliation records (uplift), forming-shop registry, custody index (sealed — metadata only), survey and cadastral*

(Council legacy). No action is required of you. Should you wish to record a purpose statement against these queries, one may be lodged at any time before signature.

Two thousand one hundred and fourteen. She had not known the number. It was, she thought, about right.

The bit that stayed with her on the taxi was not *no action is required*. It was the word *categories*. Somebody had sat with her queries and sorted them into four piles, and one of the piles was the survey and cadastral, which had nothing to do with tuning, and nobody had had to be told to look for it; the categories were just there, ready, in the form.

She read it again at Bree.

Then she lodged the purpose statement, because you may as well be the one who writes your own file. She typed it standing up, one thumb, in the queue for the rank.

Purpose: reconciling family-B guide segments (form 12, closed) against forming reference. Survey queries: personal, family papers, north-west quadrant. Happy to discuss.

She read that last part and left it in.

The road south, and the Atlantic • four weeks to Renewal

The consignment went out of Lancaster on eleven crates and one ledger seal, and the seal was the part that could not be replaced.

The crates were ash, dovetailed, lined with felt and pine shavings, built by a man in Bird-in-Hand who had built cider presses all his life and asked once what the wood was for and then did not ask again. Two hundred sixty-one kilograms of formed gold: guide segments, four collars, and the long items that the forming shop at Gap called horns and the manifest called nothing at all. Ruth had weighed every crate herself on the co-op's beam scale, because the beam scale did not need power and did not need trusting.

Her signature was on the manifest under the words RECEIVED AND CLOSED.

They went south by rail. Traction platcaps, two hundred and forty tonnes of train, charging four hours at every siding where the wind farms sat, forty kilometres an hour on the good stretches and twelve where the roadbed had been relaid over the old right-of-way by hand crews in the seventies. The bed had piezo bolts under the ties. Somebody had done the sums and the bolts paid for the signal lamps and nothing else, and everyone in the Corps knew that and everyone in the Corps still said the railway ran on its own bumping.

Beiler counted the crates at every stop. Eleven. He counted them at Perryville, at Ashland, at Rocky Mount. It is not that he doubted

the count. Counting is what you do with your hands when you cannot sleep.

The interior south of the Potomac is a country of two kinds of place.

There are the towns that came back. Brick main streets, mesh nodes on the church steeple, a market with awnings and a scale, draft teams at the rail, a diesel or two under tarpaulin for the county's emergencies. Barn raisings still going, seventy years on, because there are still barns to raise.

Then there are the substation yards.

They cut the copper out of those in the sixties and the steel in the seventies. What is left are the transformer hulks, too heavy to move and too full of oil to burn cleanly, sitting in fields of Johnson grass with their bushings gone and their tanks split where they cooked. The pylon lines were felled and taken for the steel, and what is left of them is the stubs, cut off knee-high, in rows, going over a rise and out of sight. Fifty kilometres of stubs, sometimes. Sumac between them.

Ruth watched them go by from the open door of the freight car with her boots hanging out and her cap in her lap. A private named Yoder sat down beside her and looked at the stubs.

"My grandfather says it took a minute and a half," he said.

"To burn?"

"To burn the whole east."

"It took about eleven hours," Ruth said. "The first minute and a half was the part he saw."

Yoder thought about that and then said what everybody says, which is that it seems a lot of iron to leave lying, and Ruth said the oil was the problem, and they left it there. Behind them Beiler was singing something with the words out of it.

She was six in March 2064. What she has of it is a set of small things in a fixed order. Her father coming in from the barn in the middle of the day, which he did not do. The kerosene lamp brought out of the cellar and the glass being wiped with a cloth, and the smell of the wick catching. Her mother taking the butter out of the propane cooler and

putting it in the springhouse, and the crock being too heavy for her mother, and Jonas carrying the other side.

Nothing burned in their district. That is the whole of the Plain memory of the Flare and it makes the rest of the country furious to this day. Nothing burned, because there was nothing there to burn.

The hunger years came from the outside, on foot, up the Newport Road.

They embarked at Charleston out of the deep berth the Corps keeps at North Charleston, on the *Ashley Fair*, sail-hybrid bulker, four wing-sails and a platcap bank amidships and wave-flex panels down her flanks that make her sound, in a beam sea, like a woman shifting in a chair.

Twenty-one days to Cape Town if the weather is ordinary.

The gold went into number three hold, forward corner, chocked and strapped, with the ledger seal photographed against a dated board and hashed and pushed onto the mesh before the hatch came over. After that the seal was the ship's problem and Ruth's problem and nobody else's on earth.

Watch on the hold was two hours. Ruth put herself in the rotation. Beiler told her a captain does not stand hold watch and she said she was a quartermaster, and he said that was not an answer, and she said no it wasn't, and took the midnight.

The log lives on a shelf by the ladder. Time, temperature, relative humidity, seal intact yes/no, initials.

Twenty-one days of that.

The temperature ran steady. The humidity went where the weather went. The gold did not care about either; the felt did, and the felt was hers.

Nights, the hold hums.

Not the gold. Hull noise coming down the frames, and the wave-flex skin working, and a bilge pump somewhere aft that cycles every

nineteen minutes and does not want fixing. Ruth sat on an upturned crate of nothing with the lamp turned to its low notch and her back against a stanchion, and after the first week she stopped bringing anything to read.

Her grandmother's Bible was in her kit and stayed there. The German in it is a language Ruth can hear better than she can see. On the ship she found she did not need the page. *Der HERR ist mein Hirte, mir wird nichts mangeln.* She would get four lines in and lose the fifth and go back to the start, and that was the watch, some nights, and it went quickly.

Other nights she had the coin out.

It is the one from the Ephrata pour, 2071, thick and badly struck, the rim gone soft from thirty years of thumbs. The Sunday version says *Be kind* and is what the children get. Hers has the whole of it, both sentences, the first one with the word in it that the county assemblies stopped reading aloud in about 2085 and that her grandmother, who was not Fold and never would be, said at least you know what they mean.

Ruth ran her thumb over the first sentence in the dark of number three hold, where nobody could see her do it, and put it back in her breast pocket.

The hunger years, as Ruth carries them, are not a famine. They are a queue.

The Compact granary for their district was at Ephrata and the district fed, at the end, something over eleven thousand people who had walked in from Reading and Philadelphia and one family from Trenton with a wheelbarrow. Her mother ran a soup line for four years and never once used the word charity, because it was a work rota and you were on it.

What Ruth remembers is the smell of wet wool drying in the assembly room and the sound of a hundred and forty people not talking while they ate.

In 1976 there was a hymn her mother liked. She has that wrong. It was not 1976.

Her brother's name is Jonas. He is two years older. When the Army moved their end of the district into the granary compound in the second winter, on account of the road people, somebody had to be left with the animals, and Jonas was fourteen and stayed with them, alone, for five weeks in a cold barn with a Springfield he was not permitted to load. That is not the staying that the family means when they say it, but it is where it started.

Ruth left at eighteen and enlisted, and Jonas stayed. He is on the home place at Gordonville with a wife and six and a hundred and forty acres and eight mules, and he has never once said anything to her about the Army, in twenty-three years, and that is not a coldness. He asks after her feet. His letters are one page, weather and stock and prices, and at the bottom, always, *we think of you*.

She writes back longer than he does. She has always written back longer than he does.

Day fourteen, the cook lost the ship's cat for a day and a half and had the whole after-guard turning out lockers, and it was in the sail locker having kittens, and the cook came up the ladder holding two of them like a man holding evidence. Beiler put it in the ledger of kindnesses. Yoder asked what the entry was and Beiler said, feeding a nursing mother, four hands, and Yoder said that was the cat's kindness not his, and Beiler said he was not going to argue theology with a private on a Tuesday.

Ruth laughed until she had to put her hand on the rail.

Day sixteen the wind went round and the wingsails came in and they ran four hours on the bank, and the hold got warm, and the humidity climbed nine points in a watch, and she wrote it down, and it came back.

That watch she thought about the bins at Christiana.

It was the wet fall of 2089. She was a first lieutenant and the district's movement officer, which meant she wrote the schedule that said which wagon went where and when the steel bins came empty for the next intake. That year the wheat came in at nineteen, twenty

per cent moisture, and the plan out of Corps was to bin it wet and run the forced-air driers, because the driers were new and the Corps had built them and the numbers said they would carry it.

The threshing men said stook it. Two of them, older, from Peach Bottom, said it in a room with her in it: leave it in the field on the stook, lose four per cent, keep the crop. They had a hundred and sixty years of doing that and no paper.

The plan stood. Ruth did not object to it. She scheduled it.

Bin four bridged. Wet grain does that; the crust arches over and hangs, and under the crust is a void, and the bin looks full and moves nothing. The order for the crust was poles from the top hatch, and the men on the crust were roped, and one of the ropes was tended by a man who had not been trained to tend a rope because Ruth's schedule had him tending a rope at 0400 and driving at 0600.

The crust went. Grain flows like water and weighs like grain.

Three men. Enos Zook, who was fifty-one. Christian Zook, his nephew. And a boy from Gap named Aaron Fisher who was nineteen years and four months and had been on the rota nine days.

Ruth was in the scale house forty metres away with a clipboard.

They pulled Christian out alive in the first ten minutes and he is alive now, in Kinzers, and he waves at her when she goes by.

The board of inquiry found the plan sound and the execution deficient, and named the deficiency at the level of the movement schedule, and her captain took the finding onto his own record, which was his right and which she had not asked for. She had asked for the opposite. She was reassigned to Fort Indiantown Gap and promoted eleven months later, on time.

The Fisher family came to her. That is the part. They came to the barracks in a spring wagon, the father and an aunt, and they brought bread, and they sat in the visitors' room at Indiantown Gap and told her she was welcome at the funeral and that they held nothing against her, and she said thank you, and they went home.

She did not go to the funeral. She has never been able to give an account of that.

Twenty years on she can say what she learned, and it is small and useful and does not weigh anything like the thing it came from: when the men with a hundred and sixty years and no paper are saying one thing and the driers are saying another, you stop the schedule and you take the four per cent.

Nobody has forgiven her, because nobody has ever accused her, and she has noticed that those are not the same and that she would prefer the accusation.

She wrote 0200, 24.1, 71%, seal intact, RS.

Day nineteen the wind came back and the sails went out and the ship leaned into it and Ruth stood at the rail before her watch with the coin in her fist and the Atlantic going by black.

She had a job at the far end of it. Deliver eleven crates. Get a receipt. And be able to tell the county assembly at Gordonville, when she stood up in a plank chair with her hat off, what the gold had been: what they paid, what they were offering, or what they were handing over. Three different sentences, and the manifest supported all three, and her people would want the true one.

That was what she had. Twenty-one days of humidity in a book and a question she could not answer with a scale.

She went down the ladder at four minutes to midnight and Yoder came up past her and said the humidity was falling and she said good, and he said, Ma'am, and she said, Get some sleep.

eGoli Spaceport, Gate Seven • three weeks to Renewal

The *Ashley Fair* came alongside at Cape Town on a Tuesday and the crates came up to Johannesburg on the Corps' own flatbeds under a Compact escort of four, and the whole thing was so ordinary that Ruth spent the rail leg waiting for the part where it stopped being ordinary.

Customs was a woman with a tablet and a cold. Gate Seven was a scanner arch, a weighbridge, and a queue of two.

The queue of two was because a Chilean consignment had come in the day before and somebody's forming certificate had the wrong annex number on it and now nine crates of Chilean gold sat in a bonded stall while two men argued about a footnote in the sun.

"How long," Beiler said.

"Since yesterday," said the gate clerk, cheerfully. "It'll be fine. It's page numbering."

Naledi got the liaison posting by list. It went up on the board outside the tuning house at half past six with the week's rotations, printed, because Kagiso believed in paper for anything a person might want to complain about.

Consignment US-11 (Lancaster/Gap). Berth 7C. Tuning liaison: N. Mokoena. Deputy: K. Sithole.

She read it twice. Karabo read it over her shoulder and made a noise.

“Two weeks ago you were closed,” he said.

“I know.”

“And now you’re liaison on the biggest inbound of the cycle.”

“I know.”

“So either nobody spoke to anybody, or somebody spoke to somebody and decided the safest place for you is inside the room with the door open.”

“Karabo.”

“I’m just saying you should know which it is.”

She did not know which it was. Kagiso, when she found him at the urn, said only that the Americans had asked for somebody who could read a family-B segment on sight and there were four people at the port who could and two of them were on the Chilean mess and one had a new baby. He said it while pouring, and he did not look up, and Naledi thought that was probably true and probably also not the whole shape of it.

Berth 7C, eleven o’clock. The port’s noise at that hour is the gantry crane and, under it, the vitrified bore breathing.

Nobody calls it breathing on paper. On paper it is the thermal cycle: the bore’s wall gives up heat to the shaft air on a period of about forty minutes and takes it back, and the air moves, and where the catwalks cross the throat you feel it on the back of your neck going one way and then, later, the other. Naledi had walked that catwalk twice a shift for six years. The wall is glass. Nobody has ever made glass like it and nobody has explained the wall, and it is a stanchion mount and a place people leave their coffee.

She got there before the truck and laid out.

Comparator on the bench, levelled. Two accelerometer pucks, cabled, calibration date checked against the sticker and then against the ledger because stickers lie. The reference rod in its case, family

B, 47 millimetres, kept at twenty degrees in a box that costs more than her flat. Thermal blanket. Log tablet. A jar of chalk for hands.

Then she waited eighteen minutes, which is the actual work of tuning liaison.

The truck came in with the Compact escort walking. The escort walked because a man walking beside a load can see a strap go slack and a man in a cab cannot; Naledi worked that out about four seconds after wondering why nobody had ridden.

The officer was mid-forties, plain-faced, hair cut off short under the cap, sleeves down and buttoned in thirty-one degrees. She had the manifest in a hard folder, actual paper in a plastic sleeve, and she was reading the crane's certificate plate on the way past like a woman who read plates.

"Captain Stoltzfus. Quartermaster, US consignment."

"Naledi Mokoena. Tuning liaison." They shook. The captain's hand was dry and had a callus across the top of the palm that Naledi could not place.

"You'll want the seal photographed before I break it."

"Please."

"And I'd like the weighbridge slip put against my beam-scale figures before we go further, if that's not a nuisance."

Naledi looked at the eleven crates. "You weighed them on a beam scale."

"Each one. In Lancaster."

"The weighbridge here is good to two hundred grams on twenty tonnes."

"So is a beam scale," Ruth said, "and mine hasn't got software."

There is a kind of person who says that to be difficult. Naledi watched her say it, flat, with no lift at the end, and decided she was not that kind. "Fine. Karabo, get the slip."

They broke the seal at 11:40. It came up on the ledger inside eight seconds — hash, timestamp, the tuning house's counter-signature — and Ruth stood and watched the screen for the whole eight seconds with her thumb on the manifest and then let out a breath through her nose and put the folder under her arm.

"That's mine done," she said.

"Not quite."

"No. But that's the part I could lose."

Crate four came out first because crate four had the horns in it, and Naledi wanted a horn on the bench before she touched a segment, for reasons she would have found hard to justify to Kagiso and did not have to.

The lid came off and the pine shavings came out by the handful into a bin, and there was the felt, and under the felt the long items lay in cradles cut to them.

They are 1.4 metres. They are not horns. The manifest calls them nothing at all: item, mass, purity, form reference. The forming shop at Gap calls them horns because they taper and because men name things. What they are is a tapered waveguide section with a flare at the wide end and a machined collar at the narrow, and the flare is not a curve you would arrive at by taste.

Naledi lifted one out on two hands with the chalk on her palms.

Nine and a bit kilos in the hands and it wants to keep going in the direction you started it. She set it in the cradle on the bench, adjusted, and put the puck on the collar with a smear of couplant.

"How long's the run," Ruth said.

"Twelve minutes for the thermal to sit. Then four minutes of it, then I do the same to the next one, and the next one, and there are nine, and then the collars, and then the segments, and we will both be here at eight tonight."

"Good," Ruth said, and sat down on a crate that was still full.

Here is what a calibration is, for anyone who has to sit through one.

You have a piece of gold formed to a shape. You strike it, softly, with a nylon-tipped hammer at a marked point — the same point, the same energy, the port has a jig for it — and you record what it does. What it does is ring at a fundamental and at a stack of overtones above that, and the ratios in that stack are the form's signature. Purity sets the losses. Geometry sets the ratios. If the forming shop has done its work the ratios sit inside a tolerance you can print on a card, and the card is what the counterparty's manifest specifies, and that is the entire trade: a shape that rings correctly.

You also get the noise. The bore breathes, the crane runs, a truck goes past on the perimeter road. That lives at the bottom of the trace and you learn to see through it the way you learn to see through a dirty windscreen.

Naledi struck the first horn at 11:56 and watched the comparator draw.

Fundamental clean. Second, third, fourth clean. Everything inside the card, which she had expected, because Gap's forming shop was reputable and because the beam-scale woman on the crate would not have brought her bad gold.

And at 2.31 times the fundamental, where the card says nothing at all should be, a line.

Not noise. Noise is a hedge. This was a picket: narrow, twelve decibels up out of the floor, sitting there for the whole four minutes of the run and not caring what she thought about it.

She looked at it for a while.

"Something's wrong," Ruth said.

"Why do you say that."

"Because you stopped moving."

Naledi struck the second horn.

Same picket. Same place. She did the third, and the fourth, and she stopped keeping her face arranged around the fifth because it was costing attention she wanted elsewhere.

Nine horns. Nine pickets at 2.31, within a fifth of a per cent of each other, in nine pieces of gold formed by hand in Pennsylvania by three different men on two different mandrels over eleven weeks.

Which is not possible from forming.

Forming error scatters. Nine pieces off the same mandrel will scatter across the tolerance like shot; nine pieces off two mandrels will scatter in two clusters. Nine pieces agreeing to a fifth of a per cent on a feature nobody was aiming at is not the shop's hand. It is something the shape *is*, or something the room *does*, and she had one way to tell which.

She swapped in the reference rod. Family B, 47 mil, twenty degrees, the port's own, formed in Germiston, never nearer Pennsylvania than this bench.

Struck it.

The picket came up at 2.31.

Naledi put the hammer down on the bench and left her hand flat next to it.

Ruth had been watching for two hours and had understood roughly a third of it, which was enough.

She understood that a line on a screen had appeared in a place where the woman had not been looking, and had then appeared in the same place nine more times, and had then appeared in a piece of metal that had nothing to do with Ruth's crates at all, and that this last had been worse rather than better.

She got up off the crate.

"Say the part I need."

Naledi did not look away from the comparator. "There's a feature in the response at two point three one times fundamental. It isn't in your gold and it isn't in mine. It's in the room."

"The room."

"Something in this facility is ringing at a frequency that shows up in everything I put on this bench, and it is not in the tolerance card, and it has been logged before, by me, three weeks ago, on the Chilean line." She turned her head. "That log was closed."

"Closed how."

“Reviewed and closed. Form twelve. It happens; most of what we file is nothing.”

“Is this nothing?”

Naledi said, “No,” and that was all she said, and Ruth noticed the woman’s tempo had gone from six words a second to two.

Ruth stood and looked at the trace for a while, because it was there and because she has always believed that if a thing is going to be difficult you should look at it early.

Then she asked the question she had, which was not any of the ones Naledi expected.

“Was that line in my crates when I loaded them?”

Naledi opened her mouth and shut it. “I don’t know.”

“Can you know?”

“Not from here.”

“Then here’s what I’ve got.” Ruth put the folder on the bench and opened it flat, and turned it around, and it was the beam-scale sheet: eleven crates, tare, gross, net, in pencil, in a column, with her initials against each line and the date. “Two hundred sixty-one kilograms and forty grams. Weighed twice, four days apart, in a stone barn at Gap, on a scale I cleaned myself. Formed on the ninth through the twenty-first of June. Three shops, and I can name the man for every item and the day he pulled it. The horns came off Ammon Stoltzfus’s mandrel and off Yoder’s, and Yoder’s is the older one and I know its number.”

Naledi looked at the column.

“Why are you giving me this?”

“Because you can tell me what the gold does and you can’t tell me where it’s been, and I can tell you where every gram of it has been to the hour.” Ruth tapped the sheet once with two fingers. “If the line came aboard at Charleston, that’s the ship. If it came at Cape Town, that’s the rail. If it was there in the barn in June, then it was there before your room existed for us, and we have got something else entirely. And there is a way to find out, which is that Gap struck every item on their own jig before crating, and they wrote it down,

because the annex makes them, and those sheets are in a drawer in Pennsylvania.”

There was a silence with the crane in it.

“You’d get those for me,” Naledi said.

“I’d get them for the manifest,” Ruth said. “The manifest is mine. You can read over my shoulder.”

They ran it again at four, with the crane stopped, which cost the port money and took Kagiso two phone calls and a favour.

The bore’s noise floor at 2.31 with the crane stopped is nineteen decibels lower than with the crane running. The picket did not move. It sat there twelve up, as clean as a rung bell, in an empty berth with nothing in it but two women, a bench, and eleven open crates.

Naledi wrote the run number in the log by hand, and then the ledger reference, and then a sentence, and then she read the sentence over and did not soften it.

Feature at 2.31 F reproduced on nine consignment items and on port reference rod, crane stopped, noise floor -19 dB. Not attributable to forming. Not attributable to instrument. Origin unknown.

She put her initials. Then she turned the tablet around.

“You’re not required to sign that.”

“No.”

“You’d be signing that you watched it.”

“That’s the only part I could sign.”

Ruth read it twice. Then she took the stylus and wrote, under it, in the witness field, *Present for runs 4 through 13. R. Stoltzfus, Capt., USA, quartermaster, consignment US-11*, and the time, and handed the stylus back.

Neither of them said anything about it. There was nothing to say that would not have been smaller than the thing.

Ruth went back to the Corps' rooms at Gate Seven at nine and sat on the bed and got her boots off and did not lie down.

She had a signature on a manifest that said RECEIVED AND CLOSED, and she had a receipt coming, and the plain reading of her orders was that in eleven days she would hand over eleven crates, take the paper, and go home and stand up in a plank chair at Gordonville and say what she was told the gold had been.

There was a version of the next three weeks where she did not mention the line.

It was not even dishonest. Nobody had asked her. A quartermaster's business is mass, seal, condition, and delivery, and the tuning was the host's affair, and a woman who has spent twenty-three years in a logistics corps knows exactly how much of the world runs on people not volunteering.

She got the coin out and turned it over once and did not look at it, because you do not need to look at a thing you have known with your thumb for thirty years.

The first sentence is about this. It has always been about this. It is about the specific and available choice to be the sort of person who lets a room close over a true thing because the paperwork does not require otherwise.

She put the coin back in her breast pocket and buttoned it and got her writing case out, and wrote to Gap for the jig sheets, and to Beiler with the escort roster for the week, and then, because she had the case open, one page to Jonas. Weather, and the ship's cat, and that the country here was drier than she expected and the mules were bigger.

She did not write anything about the line. She would not have known how to say it in one page.

Naledi stayed until the maintenance crew came to lock the berth and then walked out along the perimeter and did not take the taxi.

She was thinking about the reference rod, which was the part that had gone through her like cold water. Nine horns from Pennsylvania she could argue about all week. The rod was hers. The rod was

Germiston steel-town gold, formed six years ago, kept in a box she had personally logged the temperature of four hundred times, and the rod had the same picket in it, which meant the picket had been in every calibration she had ever run at this port and she had been looking straight through it for six years, the way you look through a dirty windscreen.

Twelve decibels up. Not hidden. Not subtle. Just outside the card, and everything outside the card is nothing, because that is what a card is for.

Her hands were still chalky. She wiped them on her trousers, which is against the standing instruction, and she stopped at the fence where the reclamation plant makes the air smell burnt and got her tablet out.

She did not file. Filing had been done and closed once and she had learned something from that, and the thing she had learned was not *don't file*.

She opened the survey directory instead, the personal one, and she typed the bearing she had estimated off a wall in Langlaagte with her hand held up like an amateur, and the two dog-legs, and the length of the spur as she had stepped it off the hand-drawn scale bar, and she saved it, and she gave the file a name.

2.31.

Then she went and got the taxi, because it was nine forty and she was not going to walk to Bree.

eGoli Spaceport, the tuning house • three weeks to Renewal

The Chilean footnote got fixed at eight in the morning and the whole port learned about it by nine, because the fix was that somebody in Santiago re-issued the certificate with Annex 4(b) instead of 4(a) and a nine-crate consignment sat in a bonded stall for thirty-one hours over one letter.

Karabo found this hilarious for about a day and a half.

“Thirty-one hours,” he said, “for a bracket.”

“For the wrong bracket.”

“Same bracket. Different letter.” He was filling in the humidity column for the stall log with a pen because the stall log is one of the four things at eGoli still kept on paper. “You know what the Chileans are going to do now.”

“Re-issue everything with both.”

“They’re going to put both annexes on every certificate for a year and then somebody will write a note saying you can’t cite two annexes and we’ll do it all again.”

Naledi laughed, and then she stopped laughing, because at half past nine the rotations went up on the board and her name was on Chilean re-certification.

It was not a demotion. That was the first thing she checked and the thing she kept checking for the rest of the week, the way you keep touching a tooth.

Chilean consignment CL-09, re-certification support: N. Mokoena (lead), K. Sithole.

Consignment US-11, tuning liaison: K. Sithole (acting), deputy T. Mahlangu.

Lead. She had been deputy on nothing and now she was lead on something, and re-certification support is real work — nine items, full family-C run, the annex trail rebuilt from the forming shop's sheets up. It is exactly the work she had asked for on US-11. It was two berths away from US-11 and it would take four days and she would spend those four days with her hands on Chilean gold, which had also had the picket in it three weeks ago, and which she was now being paid to certify.

Kagiso was at the urn again. He is always at the urn; she has begun to think the urn is where the tuning house does its actual business.

"You gave the American line to Karabo."

"Karabo's ready."

"Karabo's ready and I'm on Chilean."

"You're lead on Chilean." He said it the way you hand somebody a coat.

"Kagiso."

He put the cup down. He is fifty-eight and has the guild's whole hand-history in his fingers, three knuckles that don't close and a thumb that got between a collar and a cradle in 2081. "Two things are true," he said. "One. The Chilean thing needs somebody who can read family C and argue about annexes, and there are two of those in this building and one of them is me and I'm in Renewal briefings until the fourteenth. Two. Commissioner's office rang the scheduler on Monday to ask whether the liaison postings were fixed or indicative."

"That's not a no to me."

"They didn't ask about you. They asked whether the postings were fixed." He picked up the cup again. "The scheduler said indicative."

Which is true. The scheduler is not a liar.”

“Kagiso—”

“I know what you’re going to say and I’d like you to say it somewhere that isn’t the urn.”

The Commissioner’s office is in the administration block on the east side, three storeys, brown brick, built 2078 and looking it, with a lobby full of the guild’s old comparators in glass cases and a mural of the first berthing that nobody likes.

Dr Sefako Marais came out to get her himself. That was the first thing he did.

He is sixty-one, heavy through the shoulders, and he was in shirt-sleeves with his cuffs turned back twice, and there was a stack of ledger printouts on the corner of his desk fifteen centimetres deep with a paper knife lying on top of it. He moved a chair for her. He asked whether she wanted rooibos or the other thing, and when she said the other thing he made a face and said good, because the rooibos was ceremonial and eight years old.

“Mokoena. Dineo’s granddaughter.”

“Yes.”

“She surveyed the Rand Deep Levels re-entry when I was a graduate. I sat in the back of a hearing once while she took a mining lawyer apart over a bearing.” He sat. “She did it entirely in numbers. He didn’t understand he’d lost for about a week.”

“That sounds like her.”

“How is she?”

“She died in ninety-eight.”

“Ah.” He put his hand flat on the desk for a second. “I’m sorry. I did know that. I was at the Codehouse funeral for Mabaso the same month and I have the two of them filed together and wrongly.” He looked at her. “That’s not an excuse, it’s just what happened.”

She had come in with a sentence ready and now the sentence was in the wrong shape.

"You logged a feature at two point three one," he said.

"Yes."

"On the Chilean line three weeks ago and again on the American items last Thursday, and on a reference rod, with the crane stopped." He had not looked at a page to say it. "Reproduced fourteen times. Not attributable to forming. Origin unknown. An American captain witnessed it in the field and initialled, which was her right and slightly unusual."

"You've read it."

"I read everything with 'origin unknown' in it. There are about forty a year." He turned his hand over. "Nine of them are somebody's phone. Four are the reclamation plant. One year we had a beautiful one at eleven hertz that turned out to be a fan in a canteen freezer that had been running out of balance since 2091. Do you know how long that took to find?"

"No."

"Fourteen months. And the man who found it is now a section head and deserves to be." He leaned back. "So. What do you want?"

She had rehearsed this too, and this part still fit. "Instrument time. Twelve hours on the gallery array with the port at Sunday load, so I can get a bearing on it. And the sealed sixty-four telemetry, because if it's the bore or the node I need to know what a rough wake looks like and there's no other record of one."

"Right." He said it without a flinch, which frightened her more than a flinch. "The instrument time is a scheduling question and I don't own the schedule; Renewal owns the schedule until the sixth of next month, and Renewal has nine delegations wanting gallery walk-throughs and a Chinese survey party who have been promised the throat catwalk twice and denied it twice. You'd be applying for a window against those. You should apply. The form is a fourteen, and it goes through the guild-master, not me, and you should ask Kagiso to sign it rather than filing it up the chain, because a fourteen that arrives without his signature comes back down to get it and that's ten days."

She wrote 14 on her tablet.

"The sixty-four telemetry is not mine either. It's Steward-sealed, and the Accord's confidentiality annex sits on top of the seal, and the release path is a joint application — guild and Stewards, both, and then Sela's office is notified, not consulted, notified, and the notification has a thirty-day clock on it." He said it evenly. "Thirty days from the twelfth is after Renewal."

"Then waive the clock."

"I can't waive the clock. I can ask the Elder Steward to shorten it and she has refused four such requests since I've held this desk, including one from me, and she has never explained a refusal to anybody, which is either the most irritating institutional habit on this continent or the reason we still have a threshold chamber. Pick one on the day of the week."

Naledi sat with that.

"You're not saying no."

"Why would I say no?" He looked genuinely puzzled, and she believed the puzzlement. "It's an unexplained twelve-decibel feature in the medium the entire species relationship runs on. If it's real I want it on the ledger with a mechanism next to it. If it's a fan in a canteen freezer I want that on the ledger too, with the fan's serial number." He pulled a printout off the top of the stack, glanced at it, put it back. "What I'm saying is that I have twenty-one days, and in twenty-one days I have to get a volume schedule, a technique schedule, a settlement mechanism and a custody clause past nine delegations and a Chinese bloc that has spent three years arguing that Africa is holding the species' door for rent. And two of those delegations are looking for a reason to say the host is unfit."

"This isn't a reason to say that."

"It is exactly the shape of one." He said it kindly. "A discrepancy in the host's own calibration chain, logged, closed, re-logged. I could write the Chinese note myself. Three paragraphs. It would be a good note."

"So you want me to sit on it for three weeks."

"No. I want you to be right about it for three weeks and then say it in a room where it can't be used." He put his palm on the printout

stack. “The difference between those two things is enormous and it is going to feel to you, for three weeks, like no difference at all. I know that. I’ve been on your side of it. I lost that argument in 2081 and I was right and the man who beat me was wrong and I’ve since come to think he was right about the timing, which is a horrible thing to have to know.”

She looked at the stack. She could read the top sheet upside down: it was a volume table, tonnes per annum by year, out to 2130, with somebody’s pencil in the margins.

“We can be curious for twenty-seven years,” Marais said. “We can be sovereign for six weeks.”

“That’s a good line.”

“It’s a true line. Those aren’t exclusive.” He stood, which ended it. “File the fourteen. Get Kagiso to sign it. And Mokoena — if it comes back with a window in it, take the window at Sunday load and don’t tell me about it in advance, because if you tell me in advance I have to tell Sela’s office, and I’d rather find out afterwards like everybody else.”

At the door he said, “The bearing you want. Off what wall?”

“Langlaagte. An old stope wall.”

“Hm,” he said. “Is that hers?”

“It’s her handwriting.”

“Then it’ll be right,” he said. “She was never wrong about a bearing. She was wrong about people.”

She filed the fourteen on the Tuesday afternoon. Kagiso signed it in front of her, and read it first, all of it, and asked her to change one word — *bore* to *facility structure*, because a fourteen naming the bore goes to the geotechnical committee, which meets monthly and had met.

It came back on Thursday.

Not rejected. Returned for completion: item 9(c), *estimated effect on scheduled Renewal activity*, left blank, and item 11, *co-signature of*

berth authority for affected berths, unsigned. She had left 9(c) blank because she did not know the answer and had written *nil expected*, and *nil expected* is not a figure. The berth authority is a man called Pieterse who is not obstructive and was in Durban until Monday.

She re-filed on Monday with 9(c) filled in as *twelve gallery-hours, Sunday load, no berth movements affected* and Pieterse's signature under it.

It came back on Wednesday. The Sunday she had asked for was gone. A Chinese survey party had the throat catwalk from 06:00 and gallery isolation could not be guaranteed with a party in the throat, which was true, and Naledi checked it, and it was true.

The scheduler offered her the following Sunday.

The following Sunday was five days before Renewal.

Nobody had said no. She wrote it out on the back of a canteen napkin that evening, all of it, the way you write out a fault tree: office asks scheduler whether postings are indicative; scheduler answers honestly; guild-master allocates by competence; Commissioner encourages the application; form returns twice for genuine omissions; berth authority genuinely in Durban; Chinese party genuinely promised the throat twice already. Every link load-bearing. Every man in the chain doing his job well.

The tree came out clean and the answer at the bottom of it was *after signature*.

She put the napkin in her pocket and did not throw it away, and later that week she found it gone soft in the wash.

The tuning house has a technical meeting on Wednesdays at four. Twenty-two people, an urn, a screen, and Kagiso running it off a half sheet.

She had the item put on: *anomalous common-mode feature, cross-consignment, request for coordinated logging protocol*. Which was small. It was the smallest thing she could ask for: that every calibration in the house, for the next three weeks, record the amplitude at 2.31 F in a named column, so that at the end of three weeks there would be four hundred data points instead of fourteen.

It cost nothing. It cost one column in a log.

She presented it in six minutes. She had the fourteen traces up and the reference rod trace last, and she said the sentence about the card and the windscreen, which she had not meant to use in a meeting and which landed.

Then the room.

Thandi Mahlangu said the column was fine but the naming was a problem: if you name a column *2.31 anomaly* in a house log, the house log is discoverable under the Accord's audit annex, and in eighteen days there will be nine delegations' technical staff reading every discoverable document at this port, and they will find a column headed *anomaly* in four hundred calibrations and no explanation next to it, and the Commissioner will spend a day of Renewal on it that he does not have. Call it a spectral note. Log the number, not the word.

Which was reasonable. Naledi said fine, call it whatever you like.

Then Sipho Nkosi, who is a good tuner and has been on the choir line nineteen years, said he did not want to log a column he could not defend. If he writes down an amplitude at a frequency, he is asserting a measurement, and the pucks are not calibrated for narrowband work at that frequency, and if the number is soft and somebody builds an argument on four hundred soft numbers, that is on him. He would log it if the house bought narrowband channels. The house has not budgeted narrowband channels.

Which was also reasonable, and Naledi, who had been irritated for four seconds, stopped being irritated, because she had thought the same thing about the pucks at eleven o'clock the previous night and had decided the ratio was robust even if the amplitude was soft, and she had not written that down anywhere, and if she had not written it down then Sipho was right to refuse.

Then somebody asked what the mechanism was, and she said she didn't know, and there was that pause that a technical room does when it has stopped being with you.

Kagiso said: no protocol change three weeks out from Renewal; log it as a free-text note at the tuner's discretion; revisit at the first meeting in the new cycle. He said it in nine seconds. He was not unkind about

it. He wrote *revisit* on his half sheet, and she saw him write it, and she believed he would.

The item took eleven minutes of a two-hour meeting. The next item was the canteen's coffee contract and it took twenty-five.

Sipho caught her at the door with his cup still in his hand.

"You're right about the ratio," he said.

"You just said—"

"I said I won't log an amplitude. The ratio's clean. A fifth of a per cent across fourteen pieces off four mandrels. You can't get that from forming and you can't get that from mounting." He drank some of the coffee, made a face, and put the cup on the sill. "I've seen it. I've seen it for years."

Naledi stood very still. "In runs?"

"In runs. It's above the card so I've never written it down. Neither has anyone. There's nowhere to write it." He shrugged, one shoulder. "I've got two on the choir line and a daughter doing her third year at the academy, and if I stand up in a meeting eighteen days out from Renewal and say the port's calibration chain has an unexplained common mode, the sentence leaves the room without me attached to it and comes back in a Chinese technical note. Then Kagiso has to defend me, and he will, and it costs him. And in ten days he has to go into a room and get the guild's custody language into the amendment, which is the only thing at this port I actually care about."

"So you'll say nothing."

"I'll say it to you at a door." He picked the cup back up. "That's not nothing. It's just not a meeting." He looked at her, straight. "Bring me a mechanism and I'll say it in the meeting with my name first on it. I'm not brave, Naledi. I'm arithmetic. Bring me a mechanism and the arithmetic changes."

He went off down the corridor. Two others said versions of it to her that week, in a car park and over the urn: Thandi, who told her to keep every trace off the house server and on her own hardware, and gave her a spare drive out of her bag; and a young tuner called Lebo who

said nothing useful at all and left a printed 2081 committee minute on her bench with a paragraph highlighted, and never mentioned it.

Nobody in that building was a coward. That was the part she could not get comfortable against. They all had a thing they were carrying and were carrying it well, and the aggregate of twenty-two people carrying their things well was a door closing without a hand on it.

Friday she went to look at the Chilean gold, because it was hers to certify and because she had begun, in a way she did not like, to think of berths as places to be near.

Nine items, family C, formed at Rancagua. She ran the full set and the picket was there at 2.31 in all nine, exactly where the American horns had it, and she wrote it in free text at the tuner's discretion, as instructed, in the box provided, at the bottom of a certificate that was otherwise entirely clean.

Then she signed the certificate, because the gold was good. That was the true thing about it. The gold was good, the annex was now cited correctly in both brackets, and the Chileans could ship.

Karabo came past at six with two rolls and put one on the bench.

"How's the American liaison," she said.

"Fine. She writes everything down twice. She weighed the crates again after unloading."

"Of course she did."

"She asked me what the two point three one column on my log was for." He tore the roll in half without much interest. "I said ask Naledi. Was that wrong?"

"No," Naledi said. "That was exactly right."

She ate half the roll and looked at the certificate on the screen with her name at the bottom of it and the true sentence sitting in a free-text box where nobody would ever query it, and she thought about the napkin, gone to pulp in her trouser pocket, and every good man in the chain.

Then she got out her tablet and opened the file called 2.31 and added a line: *Sunday 26th. Five days out. Take it.*

74EGOLI SPACEPORT, THE TUNING HOUSE · THREE WEEKS TO RENEWAL



Klaus

Custos, non Conditor